Case Report

Removal of a Rare Foreign Body from the Right Secondary Bronchus

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Introduction:

Foreign body aspiration is an important airway emergency, particularly in paediatric age group. Children are especially vulnerable due to their tendency to place objects in their mouths, immature protective airway reflexes, and incomplete coordination between swallowing and respiration. Aspirated foreign bodies can become lodged in the larynx, trachea, or bronchial tree, based on their size, shape, and type. When they reach the bronchi, they may cause cough, wheezing, diminished airflow, respiratory distress, or sometimes only mild symptoms. Prompt diagnosis and removal are essential to prevent complications such as airway obstruction, atelectasis, pneumonia, and bronchiectasis.

Case Report:

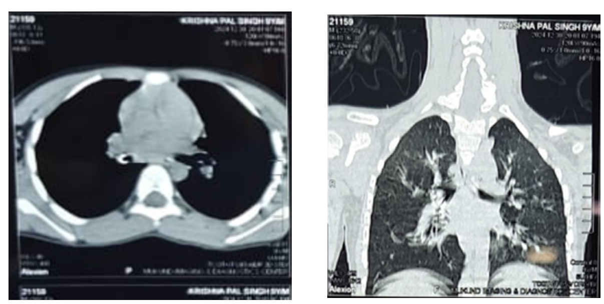
An 8-year-old male child, firstborn, resident of Ratlam, presented to emergency department with a history of accidental foreign body inhalation one day before admission. According to mother's history, the child had apparently been well until accidentally inhaling a broken whistle. Following the incident, he developed cough, difficulty in breathing, and a whistling sound during forceful expiration.

There was no significant past medical or surgical history. On general examination, child was conscious, oriented, afebrile, and in fair general condition. There was no pallor, icterus, cyanosis, lymphadenopathy, or oedema. Respiratory examination revealed an oxygen saturation of 98% on room air and a respiratory rate of 22 breaths per minute. The patient showed no signs of acute respiratory distress. However, reduced air entry was noted on the right side, accompanied by rhonchi and expiratory wheezing. Cardiovascular exam revealed normal S1 and S2 sounds with no murmurs. Abdominal and neurological examinations were unremarkable.

Investigations:

Routine blood investigations showed haemoglobin of 13.9 g%, platelet count of 4.48 lakhs, and total leukocyte count of 18,200/mm^3. Renal function tests were within normal limits, with urea/creatinine of 17/0.49 mg/dL.

Serum electrolytes were normal, with sodium 140 mEq/L, potassium 4.18 mEq/L, and chloride 105 mEq/L. Chest X-ray showed clear bilateral lung fields. However, a CT scan of the neck and chest revealed a 7 mm hyperdense cylindrical structure, suggestive of a foreign body, located in the distal right secondary bronchus approximately 2.5 cm distal to the carina. It was causing partial obstruction with air trapping in the right lower lobe. No other significant abnormality was detected on CT scan.



Based on the clinical and radiological findings, a diagnosis of non-vegetative foreign body in the right secondary bronchus was made.

Fig. 1. Foreign body seen in CT – Neck with chest

Management

The patient was planned for rigid bronchoscopic foreign body removal using optical forceps. High-risk consent, consent for possible postoperative ICU care, and ventilator consent were obtained. After pre-anaesthetic clearance, the patient was taken for surgery.

Under sedation, the child was placed in the supine posi-

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tion. A Macintosh laryngoscope was introduced into the oral cavity under direct vision, followed by insertion of a rigid bronchoscope of size 3.5 into the trachea. The carina was visualized using optical forceps. The foreign body was identified in the distal part of the right secondary bronchus.

The foreign body was removed in two pieces after two attempts. A bronchoscopy was subsequently performed to rule out any retained foreign body fragments. Thick mucopurulent pus was suctioned from the airway. Bronchoscopic images confirmed visualisation and successful removal of the foreign body.

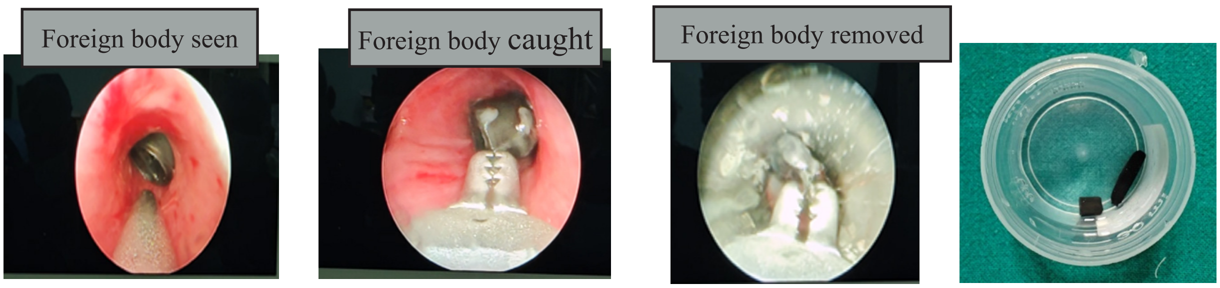

Fig.2. Steps of foreign body removal.

Postoperative Care

Postoperatively, the child was shifted to the paediatric intensive care unit for observation for one day. Intravenous antibiotics were administered to prevent pulmonary infection. Strict oxygen saturation monitoring was maintained. Nebulization with budesonide and salbutamol was given, along with chest physiotherapy. The child remained stable during the postoperative period.

Discussion

Foreign body aspiration remains a potentially life-threatening condition in children. The nature, size, and shape of the foreign body influence its site of lodgement. Larger foreign bodies may become impacted at the level of the glottis or sub glottis, while smaller objects may pass into the trachea or bronchial tree. Sharp foreign bodies such as pins, needles, and fish bones can become lodged anywhere in the laryngo-tracheo-bronchial tree.

Children are at increased risk because they commonly place objects in the mouth and may cry, walk, or run while holding objects orally. In addition, lack of molars, immature swallowing coordination, elevated laryngeal position, and underdeveloped protective reflexes contribute to the risk of aspiration. In adults, aspiration may occur during coma, deep sleep, alcohol intoxication, anaesthesia, or due to loose teeth and dentures.

The most common symptoms of airway foreign body aspiration include choking, cough, wheeze, and decreased breath sounds. In the present case, the child had cough, breathing difficulty, a whistling sound on expiration, reduced right-sided air entry, and wheeze. Although the chest X-ray was normal, the CT scan clearly demonstrated the foreign body in the right secondary bronchus with associated air trapping. This highlights the importance of advanced imaging when clinical suspicion remains high despite a normal chest radiograph.

The gold standard for diagnosing and treating pediatric tracheobronchial foreign bodies is still rigid bronchoscopy. It makes safe foreign object extraction, direct viewing, and airway management easier. In this instance, the broken whistle pieces were successfully removed using rigid bronchoscopy and optical forceps. after which a check bronchoscopy is performed to guarantee full clearance.

Conclusion

Foreign body aspiration should be suspected in any child presenting with a sudden onset cough, wheeze, unilateral reduced air entry, or abnormal respiratory sounds following a choking episode. A normal chest X-ray does not exclude the diagnosis. CT scan can help localise radiopaque or suspected foreign bodies, especially when symptoms persist.

The best course of action is still early intervention with rigid bronchoscopy. Preventing morbidity and guaranteeing positive results requires prompt diagnosis, cautious airway management, total removal, and suitable postoperative care.

Reference:

1. Kryger M, Bode F, Antic R, Anthonisen N. Diagnosis of obstruction of the upper and central airways. Am J Med. 1976;61(1):85–93. doi: 10.1016/0002-9343(76)90048-6. [DOI] [PubMed] [Google Scholar]
2. Jackson C, Jackson CL. Bronchoesophagology. WB Sauders; Philadelphia: 1950. Bronchoscopy; pp. 50–67. [Google Scholar]
3. Tietjen PA, Kaner RJ, Quinn CE. Aspiration emergencies. Clin Chest Med. 1994;15(1):117–135. [PubMed] [Google Scholar]
4. Heimlich HJ. A life-saving manoeuvre to prevent food choking. JAMA. 1975;234(4):398–401. [PubMed] [Google Scholar]
5. Kim IG, Brummitt WM, Humphry A, Siomara SW, Wallace WB. Foreign body in the airway: a review of 202 cases. Laryngoscope. 1973;83(3):347–354. doi: 10.1288/00005537-197303000-00004. [DOI] [PubMed] [Google Scholar]